Macroeconomics

Time: 120 minutes

Total: 10 points

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Backup Exam

This exam contains two parts.

Using AI is strictly prohibited. Reviewing past lecture notes are possible.

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1. (4 points) Consider the following dynamic optimization problem

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

where

$$u(c_t) = ac_t - \frac{b}{2}c_t^2, \qquad c_t < \frac{a}{b} \forall t$$

subject to

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t.$$

given $\beta, \delta \in (0,1)$ and $a, b > 0$.

Answer the following questions

(a) Write the Lagrangian and derive the Euler equation of the optimal consumption-saving.

(b) Compute the intertemporal elasticity of substitution $IES = -\dfrac{u'(c)}{c\,u''(c)}$. Explain the meaning and its implications for a poor household with low c versus a rich household with high c.

(c) Derive the change in consumption $c_{t+1} - c_t$. Assume the economy is away from the steady state so that $\beta R_{t+1} \neq 1$. What does a higher b imply for the consumption smoothing behavior when $\beta R_{t+1} \neq 1$? Derive the steady state level of consumption c^*. Assume that $f(k) = Ak^\alpha$. What is the steady state level of capital k^*?

(d) Assume that the government makes a one-time positive transfer to households at $t = 0$. What would be the likely response of poor household (low c) and rich household (high c) regarding consumption now (and saving for later)?

2. (6 points) We want to study the role of polygyny (đa thê: một người đàn ông có thể lấy nhiều người phụ nữ làm vợ) and bride prices on capital accumulation and the macroeconomy.

Production

A representative firm produces output Y_t using capital K_t and labor L_t with a Cobb–Douglas technology:

$$Y_t = AK_t^{\alpha} L_t^{1-\alpha}, \qquad A > 0,\ 0 < \alpha < 1. \tag{1}$$

Capital depreciates fully within one period ($\delta = 1$). Each young worker supplies one unit of labor inelastically, so $L_t = N_t$. Define $k_t \equiv K_t/N_t$.

Demography

Time is discrete, $t = 0, 1, 2, \ldots$ In each period N_t identical young males are born; each lives for two periods—young (period t) and old (period $t+1$). The population of young males evolves according to

$$N_{t+1} = n_t N_t, \tag{2}$$

where n_t is the number of children per male, chosen endogenously.

Household

A male household head may take f_t wives at a **bride price** $p > 0$ per wife. Each wife helps with child-rearing: the total resource cost of raising n_t children with f_t wives is n_t^2/f_t (convex in n_t, decreasing in f_t).

The male maximizes lifetime utility

$$U_t = \ln c_t^y + \beta \ln c_{t+1}^o + \gamma \ln n_t, \qquad \beta, \gamma > 0, \tag{3}$$

subject to

$$c_t^y + s_t + p f_t + \frac{n_t^2}{f_t} = w_t, \tag{4}$$

$$c_{t+1}^o = R_{t+1} s_t, \tag{5}$$

where $s_t \geq 0$ is saving and R_{t+1} is the gross return on capital.

Capital accumulation. $K_{t+1} = N_t s_t$.

Answer the following questions

(a) Write the firm's profit maximization problem and derive R_t and w_t.

(b) Solve the household problem for the optimal c_t^y, s_t, n_t, and f_t.

(c) Derive the law of motion for capital per worker, k_{t+1}. What does the absence of k_t in the expression imply about the transitional dynamics?

(d) What condition determines whether the economy is polygynous ($f_t > 1$) or monogamous ($f_t \leq 1$)?

(e) Find the steady-state values k^*, n^*, and f^*.

(f) When the bride price p increases, fertility n^* falls. What happens to steady-state capital per worker k^* and the number of wives f^*? Explain the intuition.

END OF EXAM
